

CONNOR LINDSELL

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PROFESSIONAL SUMMARY

Final-year Mechanical and Mechatronic Engineering student with practical experience delivering mechanical design and automation projects in a live aluminium manufacturing environment. Experienced across the full engineering lifecycle, including concept development, CAD, FEA, design verification, manufacturing documentation, supplier engagement, installation and commissioning support. Combines strong mechanical design capability with controls, robotics and programming knowledge to develop safe, data-driven solutions that improve reliability, productivity and maintainability.

TECHNICAL EXPERIENCE

Undergraduate Engineer (Mechanical & Mechatronic)

Dec 2025 – Present

Capral Aluminium, Smithfield, NSW

- Deliver mechanical design and automation projects in a live aluminium extrusion environment, managing work from concept through implementation using data-driven design decisions.

Hot Line Billet Roller System Upgrade

- Analysed production and maintenance data to define a billet-handling improvement opportunity associated with a 1.17% scrap rate, 69.95 tonnes of billet scrap and 42.53 hours of project-relevant downtime in 2025.
- Redesigned the roller and transfer system by reducing roller centres 41.2% and the maximum transfer 42.3%, increasing billet support by 70% and enabling stable handling of billets down to 228 mm; also simplified the mechanism by eliminating the maintenance heavy and unreliable anvil and replacing fault prone hydraulic cylinders with pneumatic cylinders resulting not only a mechanical improvement but also in safety.
- Delivered Autodesk Inventor models, GD&T-compliant drawings, static structural and thermal FEA in Inventor Nastran, design calculations and a value analysis to justify expenses.
- Coordinated suppliers, toolmakers and mechanical fitters through manufacture and installation planning whilst remaining under budget.

Packing Line Extension and Automation Upgrade

- Designed and engineered mechanical assemblies using Autodesk Inventor, applying FEA to lifting mechanisms and structural members, delivering a projected 50 percent increase in line productivity.
- Performed load calculations to support motor selection, pneumatic sizing, bearing validation, and sensor integration, implementing positional height feedback and zone-based safety limit switches.
- Produced GD&T-compliant manufacturing drawings, negotiated with suppliers, and currently overseeing fabrication, QA inspections, and PLC ladder logic integration within existing Rockwell control architecture.

Disability Support Worker

Apr 2025 – Feb 2025

What Ability

- Support neurodiverse clients through tailored activity planning and real-time adaptive problem solving in dynamic, unscripted environments.
- Demonstrate independent decision-making, emotional intelligence, and calm leadership under unpredictable conditions.

Office Manager

Mar 2025 – Oct 2025

Essential Health Chiropractic, North Kellyville, NSW

- Implemented process improvements and software automation to enhance operational efficiency and workflow consistency.
- Coordinated team operations and stakeholder communication, strengthening accountability and service performance.

EDUCATION

Bachelor of Engineering (Honours)

Feb 2022 – Nov 2026

University of Technology Sydney (UTS)

- Major in Mechanical and Mechatronic Engineering; Distinction average

TECHNICAL SKILLS

Design & Simulation: Autodesk Inventor, Inventor Nastran, SolidWorks, Fusion 360, FEA (Static Structural), Topology Optimisation (SolidWorks & Inventor), ANSYS Fluent (CFD)

Robotics & AI: ROS 2 & ROS 1, Gazebo (Ignition), RViz, OpenCV, RGB & Depth Cameras, RL, LfD, Hierarchical Control System

Mechanical & Manufacturing: GD&T, DFM / DFA, Sheet Metal Design, 3D Printing, Laser Cutting, Load Validation

Tools & Languages: C++/C, Python, GitHub, VS Code, MATLAB, Arduino (Uno, Mega, ESP32), Microsoft 365 (Teams, Word, Excel, OneDrive)

TECHNICAL PROJECTS

WordleBot – Autonomous Wordle Playing Robot

Mar 2026 – May 2026

Robotic Studio 2 | UTS

- Designed and trained a MaskablePPO reinforcement-learning policy to sequence pick-and-place operations across a Wordle game board, using action masking, curriculum learning and reward shaping to achieve a 100% success rate across 20 evaluation episodes.
- Outperformed a deterministic nearest-neighbour baseline by reducing average robot travel distance by 5%, demonstrating globally optimised task sequencing rather than locally greedy motion selection.
- Developed and integrated the ROS 2 and MoveIt 2 low-level control architecture for a UR3e and OnRobot RG2 gripper, implementing collision-aware motion planning, attached-object handling, trajectory scoring, collision recovery and mission control.

Pliers Design Challenge for Additive Manufacturing

Apr 2025 – May 2025

Additive Manufacturing | UTS

- Applied topology optimisation in SolidWorks, reducing effective weight by 22% (35.2 g to 27.6 g) while increasing load capacity by 82% (64.6 N to 117.5 N) through iterative prototyping and testing of six designs.
- Refined optimised meshes in Blender for printability, then tuned Creality Print 6 slicer settings, including infill patterns, support strategies and print parameters, to balance weight reduction with functional strength.
- Conducted experimental testing using a force meter to measure load at failure, documented fracture locations and validated optimisation outcomes under representative mechanical loading.

VOLUNTEERING

Project Protect

July 2024 – Present

Ski Racing Australia (SRA)

- Collaborate with a multidisciplinary team to research, design and prototype alternative safety solutions.
- Engage with stakeholders, industry professionals and academics in neuroscience and engineering to inform evidence-based design decisions.
- Develop safety equipment and protocols addressing spinal, nerve and concussion-related injuries.

TRANSFERABLE SKILLS

Problem Solving

- Analyse and resolve complex technical issues under pressure while maintaining operational continuity.

Rapid Learning

- Rapidly learn new tools, technologies and systems, applying new knowledge to improve engineering performance and workflow.

Leadership

- Led cross-functional teams of up to 50 staff, applying project coordination, conflict resolution and time-management skills to maintain performance and motivate team members.

REFERENCES

Available upon request